Meeting 5 of 7 · November 10, 2026

POLICY SELECTION

Does choosing mitigation need a learned controller?

Learned-controller hypothesis · 64-model pool · label-free probes · categorical and continuous noise · gates

Phases 5A–5C (completed Oct 7–8, 2026). Two pre-registered gates found no material structure left for a learned controller, so none was trained. A two-threshold probe rule emerged instead.

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Proposed split: Mayank (slides 1–3) · Tushar (slides 4–5) · about 5 minutes



The decision we proposed to learn

Context
model · noise · shot budget · batch

→

Label-free probes
measured every batch

→

Policy
learned controller or simple rule

→

Action A0–A4
none · readout · ZNE · both · 3× shots

H-control (original)

A learned controller (Q-TRACE) using only label-free probes beats fixed policies and a fairly tuned threshold rule.

The alternative we committed to report

A simple threshold rule matches the learned controller. Formalized for the sealed test as: the rule is near the cost-aware Oracle.

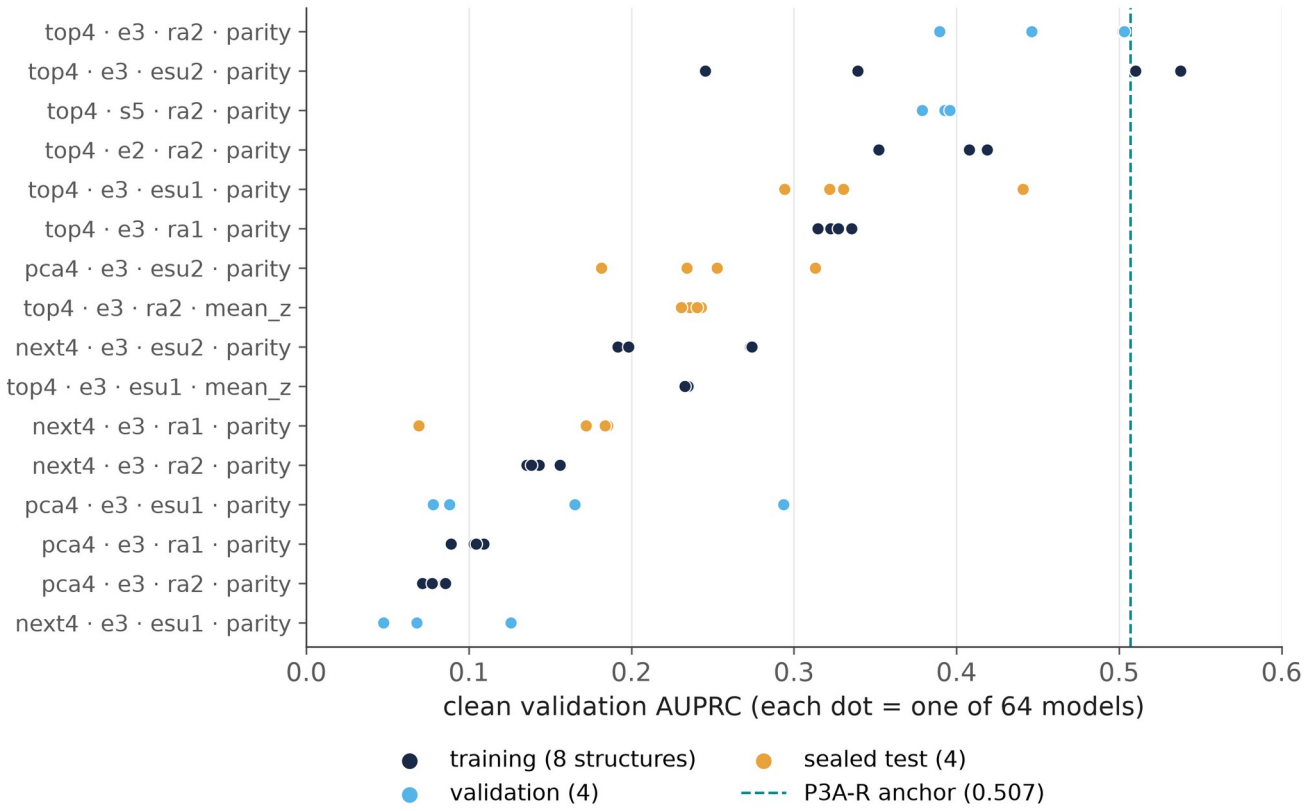
Oracle and failure (evaluation only)

- Oracle: cheapest action within 10% of the best Brier
- Failure: any disagreement with the Oracle
- Material: Brier > 1.2 × best, or ≥ 1 extra shot-set

Registered rule: train a controller only if a gate finds material structure that a simple rule misses.

Source: docs/PREREG_P5B.md, docs/PREREG_P5C.md, configs/mixed_noise_study/p5c_v1.yaml

64 independent models and two cheap probes



Readout probe

The 2 calibration circuits readout mitigation already needs. Tracks the true mean readout error: Spearman ρ = 0.999.

Gate probe

32 fixed synthetic inputs at fold factors 1 and 3. Contraction tracks depolarizing noise (ρ = −0.89) but barely thermal noise (ρ = −0.12).

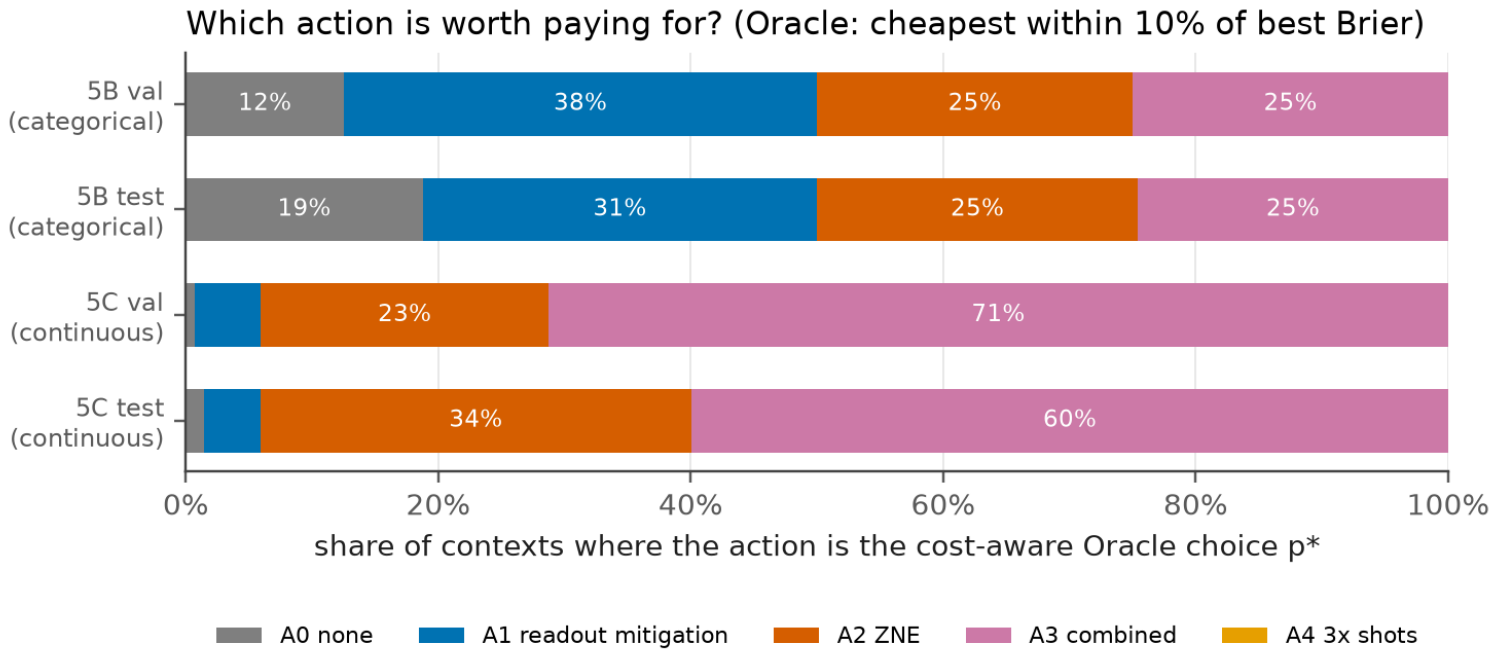
Cost and inputs

66 circuits (66·S shots) per batch · 16 static descriptors + 11 probe features

True noise parameters, labels and identifiers never enter a policy, as enforced by a feature registry and tests.

Source: p5a_pool_metrics.csv, p5b_split.csv, p5b_feature_definitions.csv, p5c_probe_validity.csv

Categorical, then continuous joint noise



Phase 5B design

8 near-categorical conditions · 1,024 and 4,096 shots · 20 replicates

Phase 5C design

17 gate settings (one gate-free) × 4 readout settings = 68 points · 256–4,096 shots

0.33%

5B: category rule wrong (validation)

67–71%

5C: combined mitigation is the Oracle's choice

< 3%

contexts where the shot budget changes the choice

0%

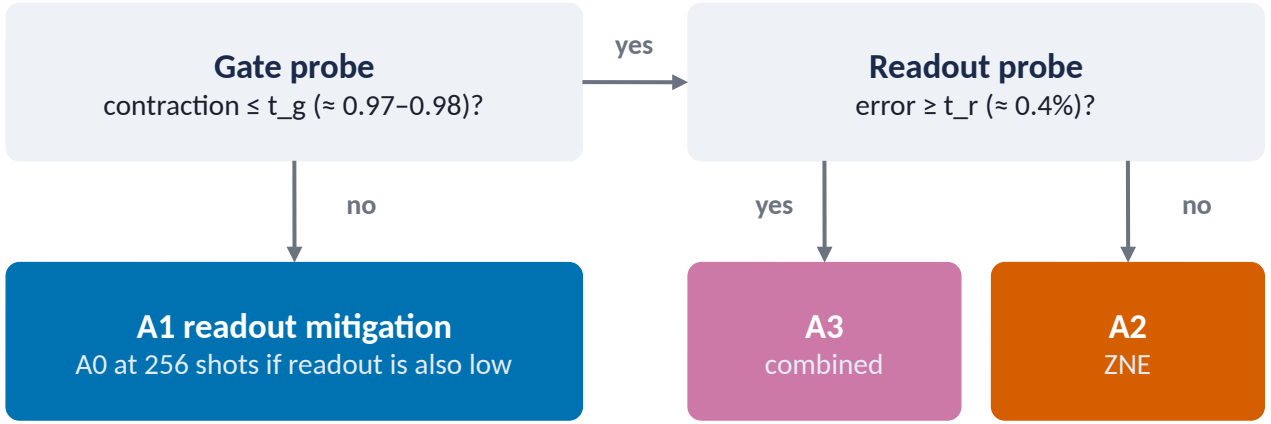
contexts where 3× shots is the Oracle's choice

Test bars come from the one-time sealed test (the Nov 24 topic).

Source: p5b_metadata.json, p5c_metadata.json, p5c_budget_effect.csv, p5c_design.csv, final_metadata.json

Both gates failed, and a simple rule emerged

The probe rule R_thr (fitted on training models only)



One readout-probe and one gate-probe threshold per shot budget. Thresholds are specific to this simulated 4-qubit setting.

Bias, not variance

shots cannot fix it, so the budget rarely matters

One question decides

in our design: is gate noise present?

G3B · categorical noise → FAIL

4-cell category rule wrong in 0.33% of validation cases (gate threshold: 5%).

G3C · continuous joint noise → FAIL

Best rule: 9.6% disagreement, but only 2.0% [1.6, 2.7] material (threshold: 5%); Brier regret +0.3%.

Residual structure

Detectable (mutual information 0.08 and 0.04 nats), mostly mean-Z readouts, but not material.

As registered, no learned controller was trained. Fixed learned comparators were evaluated on the sealed test instead (run Oct 8; Nov 24 topic).

Source: p5b_metadata.json (g3b), p5c_metadata.json (g3c, R_thr cells and thresholds, structure association), p5c_policy_eval.csv